

MATH4996 Trinomial Tree & Richardson Extrapolation for Option Pricing

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Abstract

This report presents a Trinomial Tree approach for pricing European and American options under the Black-Scholes framework. The study examines the convergence behavior of the Trinomial Tree method with respect to the number of time steps N , analyzing pricing accuracy and computational time. Richardson Extrapolation is employed as a benchmark for error estimation, avoiding circular reasoning associated with direct comparison to the Black-Scholes formula. Results demonstrate second-order convergence, with errors decreasing quadratically as N doubles, and highlight the efficiency gains from extrapolation techniques.

1 Methodology

1.1 Trinomial Tree Model

The Trinomial Tree approximates the geometric Brownian motion of the underlying asset price by allowing three possible movements per time step: upward (u), middle (m), or downward (d). This model matches the mean and variance of the log-normal process while introducing a middle branch for improved stability and convergence.

Key parameters include:

- Time step size: $\Delta t = T/N$
- Drift adjustment: $\nu = r - 0.5\sigma^2$
- Step size: $\Delta x = \lambda\sigma\sqrt{\Delta t}$, with $\lambda = \sqrt{3}$ for kurtosis matching
- Transition probabilities: p_u, p_m, p_d solved from the system ensuring mean matching, variance matching, and $p_u + p_m + p_d = 1$

The probabilities are computed as:

$$p_u = \frac{1}{2} \left(\frac{V}{\Delta x^2} + \frac{\nu \Delta t}{\Delta x} \right), \quad p_d = \frac{1}{2} \left(\frac{V}{\Delta x^2} - \frac{\nu \Delta t}{\Delta x} \right), \quad p_m = 1 - p_u - p_d,$$

where $V = \sigma^2 \Delta t + \nu^2 (\Delta t)^2$.

Option values are computed via backward induction on the recombining lattice, with terminal payoffs $\max(S_T - K, 0)$ for calls and $\max(K - S_T, 0)$ for puts. For American options, early exercise is checked at each node: $\max(\text{intrinsic value}, \text{continuation value})$.

1.2 Richardson Extrapolation for Convergence Acceleration

To estimate the true option price without relying on the analytical Black-Scholes formula (which would introduce circularity in validating the numerical method), Richardson Extrapolation (RE) is applied to sequences of Trinomial prices $P(N)$ for dyadically increasing $N = 2^k$. Assuming second-order convergence $P(N) = P_0 + c/N^2 + O(1/N^3)$, the first-order RE is: $P(N, 2N) = \frac{4P(2N) - P(N)}{3}$.

1.3 Theoretical Verification

A hardcoded test case validates the implementation:

- Parameters: $S = 60$, $T = 1.0$, $r = 0.03$, $\sigma = 0.3$, $K = 65$
- Compares convergence patterns against expected second-order behavior

Table 1: Convergence Analysis with Richardson Benchmarks

(a) European Call					(b) European Put				
N	Price	Abs Err	Pct Err (%)	Time	N	Price	Abs Err	Pct Err (%)	Time
64	5.901000	0.001891	0.0321	0.01	64	8.979960	0.001891	0.0211	0.00
128	5.901377	0.002268	0.0385	0.04	128	8.980337	0.002269	0.0253	0.02
256	5.902716	0.003607	0.0611	0.15	256	8.981676	0.003607	0.0402	0.07
512	5.901014	0.001905	0.0323	0.98	512	8.979974	0.001905	0.0212	0.33
1024	5.898205	0.000904	0.0153	3.39	1024	8.977165	0.000904	0.0101	1.27
2048	5.898537	0.000572	0.0097	8.57	2048	8.977497	0.000572	0.0064	5.42
4096	5.899147	0.000038	0.0006	25.25	4096	8.978107	0.000038	0.0004	24.39
8192	5.899119	0.000010	0.0002	99.03	8192	8.978078	0.000010	0.0001	99.45
Benchmark=5.899109, Std. Err.=0.000567					Benchmark=8.978069, Std. Err.=0.000567				
(c) American Call					(d) American Put				
N	Price	Abs Err	Pct Err (%)	Time	N	Price	Abs Err	Pct Err (%)	Time
64	5.901000	0.001891	0.0321	0.01	64	9.256062	0.002547	0.0275	0.01
128	5.901377	0.002268	0.0385	0.02	128	9.257189	0.001421	0.0153	0.02
256	5.902716	0.003607	0.0611	0.09	256	9.260147	0.001537	0.0166	0.10
512	5.901014	0.001905	0.0323	0.41	512	9.259444	0.000834	0.0090	0.41
1024	5.898205	0.000904	0.0153	1.68	1024	9.257578	0.001032	0.0112	1.67
2048	5.898537	0.000572	0.0097	8.23	2048	9.258029	0.000581	0.0063	8.46
4096	5.899147	0.000038	0.0006	32.19	4096	9.258577	0.000032	0.0004	32.11
8192	5.899119	0.000010	0.0002	132.75	8192	9.258602	0.000008	0.0001	131.33
Benchmark=5.899109, Std. Err.=0.000567					Benchmark=9.258610, Std. Err.=0.000456				

2 Analysis and Discussion

2.1 Convergence Patterns and Error Analysis

Absolute errors decrease approximately by a factor of 2 when N doubles, consistent with the $O(1/N^2)$ error term for recombining lattice methods. For instance, in the European Call (Table 1), the absolute error drops from 0.003607 at $N = 256$ to 0.000904 at $N = 1024$, a ~ 4.0 -fold reduction, and further to 0.000010 at $N = 8192$.

Percentage errors follow suit, shrinking from $\sim 0.06\%$ to $< 0.001\%$, with standard errors stabilizing below 0.001, indicating reliable extrapolation. American options exhibit similar behavior, though puts show slightly higher errors due to early exercise premiums, which introduce non-smoothness in the payoff.

Computational time scales cubically ($O(N^3)$ due to the $O(N^2)$ nodes and $O(N)$ backward steps), rising from 0.01s at $N = 64$ to ~ 100 s at $N = 8192$, underscoring the need for acceleration techniques like RE.

2.2 Advantages of Richardson Extrapolation as Benchmark

Direct comparison to the Black-Scholes formula is avoided to prevent circular reasoning: the Trinomial Tree is designed to converge to Black-Scholes, so using it as a benchmark would tautologically validate the method without assessing numerical fidelity. Instead, RE leverages the asymptotic error expansion $P(N) = P_0 + c/N^\gamma + O(1/N^{\gamma+1})$ (with $\gamma = 2$) to compute higher-order approximations from coarse grids.

Table 2: Convergence Analysis with Higher-Order Richardson Extrapolation

(a) European Call					(b) European Put				
N	Price	Abs Err	Pct Err (%)	Time	N	Price	Abs Err	Pct Err (%)	Time
64	5.901000	0.001972	0.0334	0.00	64	8.979960	0.001972	0.0220	0.00
128	5.901377	0.002349	0.0398	0.02	128	8.980337	0.002349	0.0262	0.02
256	5.902716	0.003688	0.0625	0.08	256	8.981676	0.003688	0.0411	0.07
512	5.901014	0.001986	0.0337	0.31	512	8.979974	0.001986	0.0221	0.31
1024	5.898205	0.000823	0.0140	1.20	1024	8.977165	0.000823	0.0092	1.25
2048	5.898537	0.000491	0.0083	5.28	2048	8.977497	0.000491	0.0055	6.54
4096	5.899147	0.000119	0.0020	23.83	4096	8.978107	0.000119	0.0013	26.17
8192	5.899119	0.000090	0.0015	97.95	8192	8.978078	0.000090	0.0010	103.16
Richardson: 2nd-order ($2^{11}, 2^{12}$)=5.899351; ($2^{12}, 2^{13}$)=5.899109; 3rd-order=5.899028. Benchmark=5.899028, Std. Err.=0.000567					Richardson: 2nd-order ($2^{11}, 2^{12}$)=8.978310; ($2^{12}, 2^{13}$)=8.978069; 3rd-order=8.977988. Benchmark=8.977988, Std. Err.=0.000567				
(c) American Call					(d) American Put				
N	Price	Abs Err	Pct Err (%)	Time	N	Price	Abs Err	Pct Err (%)	Time
64	5.901000	0.001972	0.0334	0.01	64	9.256062	0.002497	0.0270	0.01
128	5.901377	0.002349	0.0398	0.02	128	9.257189	0.001371	0.0148	0.02
256	5.902716	0.003688	0.0625	0.10	256	9.260147	0.001587	0.0171	0.10
512	5.901014	0.001986	0.0337	0.42	512	9.259444	0.000884	0.0095	0.41
1024	5.898205	0.000823	0.0140	1.68	1024	9.257578	0.000982	0.0106	1.67
2048	5.898537	0.000491	0.0083	8.39	2048	9.258029	0.000531	0.0057	8.56
4096	5.899147	0.000119	0.0020	32.70	4096	9.258577	0.000018	0.0002	32.43
8192	5.899119	0.000090	0.0015	134.01	8192	9.258602	0.000042	0.0005	132.99
Richardson: 2nd-order ($2^{11}, 2^{12}$)=5.899351; ($2^{12}, 2^{13}$)=5.899109; 3rd-order=5.899028. Benchmark=5.899028, Std. Err.=0.000567					Richardson: 2nd-order ($2^{11}, 2^{12}$)=9.258760; ($2^{12}, 2^{13}$)=9.258610; 3rd-order=9.258560. Benchmark=9.258560, Std. Err.=0.000456				

Wikipedia notes RE's generality for improving finite-difference methods by eliminating low-order errors via linear combinations, enhancing efficiency without additional structural assumptions. This self-consistent approach confirms the Trinomial's theoretical convergence rate while quantifying practical improvements.

3 Conclusion

This study validates the Trinomial Tree's efficacy for European and American option pricing, demonstrating quadratic error reduction with increasing N and manageable computational scaling. Richardson Extrapolation provides a robust, non-circular benchmark, accelerating convergence and enabling precise error quantification. The method's superior stability over binomial lattices, combined with RE, positions it as a practical tool for financial modeling. Future extensions could incorporate stochastic volatility or barrier options.

References

- [1] Zhou, Y., & Wang, Y. "Accelerating Convergence in Trinomial Option Pricing: Recursive Incremental Value Ordering with Repeated Richardson Extrapolation." <https://www.iieta.org/journals/mmep/paper/10.18280/mmep.100631>
- [2] Doga, L. "Heston and Trinomial Tree Pricing Models." <https://github.com/lucadoga/Heston-and-Trinomial-tree-pricing-models/blob/main/Trinomial%20Trees%20and%20Heston%20model.ipynb>

A Theoretical Foundations of Richardson Extrapolation

A.1 Overview and Motivation

Richardson Extrapolation (RE) is a sequence acceleration technique that improves the accuracy of numerical approximations by combining estimates from successively refined discretizations. It assumes an asymptotic error expansion and cancels leading error terms via linear combinations, yielding higher-order approximations without solving the full problem at finer grids. As per the referenced study, RE addresses the Trinomial Tree's slow convergence ($O(1/N^2)$) by recursively eliminating error coefficients. This is particularly valuable for nested discrete models like trees, where direct analytical benchmarks (e.g., Black-Scholes) risk circular validation. Wikipedia emphasizes RE's broad applicability to integration, differentiation, and PDE solvers, noting its simplicity: no prior knowledge of error constants is needed beyond the order γ . For Trinomial pricing, with $\gamma = 2$, it transforms coarse solutions into near-exact estimates, enhancing efficiency in high-dimensional or real-time applications.

A.2 Mathematical Framework

Consider an approximation $P(h) \approx P_0$ for step size $h > 0$, with expansion: $P(h) = P_0 + \sum_{k=1}^m a_k h^{\gamma_k} + O(h^{\gamma_{m+1}})$, where $0 < \gamma_1 < \gamma_2 < \dots$. For $h_j = h_0 \omega^j$ ($\omega \in (0, 1)$), RE computes: $P(h_j, h_{j+1}) = \frac{P(h_{j+1}) - \omega^{\gamma_1} P(h_j)}{1 - \omega^{\gamma_1}}$, eliminating the γ_1 term, yielding $O(h^{\gamma_2})$ error. Recursion extends to higher orders: $P_n^{(j)} = \frac{P_{n-1}^{(j+1)} - c_n P_{n-1}^{(j)}}{1 - c_n}$, with $c_n = \omega^{\gamma_n}$. In this work, dyadic $N = 2^k$ implies $h \propto 1/N$, and second-order RE provides the benchmark, with third-order for refined analysis.

Appendix: Code Implementation

```
# -*- coding: utf-8 -*-
"""MATH4999-TrinomialTree-Finalized.ipynb

Automatically generated by Colab.

Original file is located at
https://colab.research.google.com/drive/13V0Mm1Xe2XIMWJ5pi-bic04QKwkVKwDG
"""

import numpy as np
from scipy.stats import norm
import time

def TrinomialProbs(r, sigma, deltaT, lambda_val=np.sqrt(3)):
    """
    Calculate trinomial tree probabilities using optimal lambda for kurtosis matching
    Solves: 1) Mean matching, 2) Variance matching, 3) Probabilities sum to 1
    """
    nu = r - 0.5 * sigma**2
    deltaX = lambda_val * sigma * np.sqrt(deltaT)

    # System of equations solution
    V = sigma**2 * deltaT + nu**2 * deltaT**2

    p_u = 0.5 * (V / deltaX**2 + nu * deltaT / deltaX)
    p_d = 0.5 * (V / deltaX**2 - nu * deltaT / deltaX)
    p_m = 1 - p_u - p_d

    return p_u, p_m, p_d, deltaX, lambda_val

def TrinomialEuCall(S0, K, r, sigma, T, N, lambda_val=np.sqrt(3)):
    start_time = time.time()

    deltaT = T/N
    p_u, p_m, p_d, deltaX, lambda_used = TrinomialProbs(r, sigma, deltaT, lambda_val)
    discount = np.exp(-r * deltaT)

    # Set up S values (at maturity)
    Svals = np.zeros(2*N+1)
    Svals[0] = S0 * np.exp(N * deltaX)
    exp_dX = np.exp(-deltaX)

    for j in range(1, 2*N+1):
        Svals[j] = exp_dX * Svals[j-1]

    # Set up lattice and terminal values
    Cvals = np.zeros((2*N+1, N+1))
```

```

    for k in range(0, 2*N+1):
        Cvals[k, N] = max(Svals[k] - K, 0)

    for j in range(N-1, -1, -1):
        for i in range(N-j, N+j+1):
            Cvals[i, j] = discount * (p_u * Cvals[i-1, j+1] + p_m * Cvals[i, j+1] + p_d * Cvals[i+1, j+1])

    price = Cvals[N, 0]
    computation_time = time.time() - start_time

    return price, computation_time

def TrinomialEuPut(S0, K, r, sigma, T, N, lambda_val=np.sqrt(3)):
    start_time = time.time()

    deltaT = T/N
    p_u, p_m, p_d, deltaX, lambda_used = TrinomialProbs(r, sigma, deltaT, lambda_val)
    discount = np.exp(-r * deltaT)

    # Set up S values (at maturity)
    Svals = np.zeros(2*N+1)
    Svals[0] = S0 * np.exp(N * deltaX)
    exp_dX = np.exp(-deltaX)

    for j in range(1, 2*N+1):
        Svals[j] = exp_dX * Svals[j-1]

    # Set up lattice and terminal values
    Pvals = np.zeros((2*N+1, N+1))

    for m in range(0, 2*N+1):
        Pvals[m, N] = max(K - Svals[m], 0)

    for j in range(N-1, -1, -1):
        for i in range(N-j, N+j+1):
            Pvals[i, j] = discount * (p_u * Pvals[i-1, j+1] + p_m * Pvals[i, j+1] + p_d * Pvals[i+1, j+1])

    price = Pvals[N, 0]
    computation_time = time.time() - start_time

    return price, computation_time

def TrinomialAmPut(S0, K, r, sigma, T, N, lambda_val=np.sqrt(3)):
    start_time = time.time()

    deltaT = T/N
    p_u, p_m, p_d, deltaX, lambda_used = TrinomialProbs(r, sigma, deltaT, lambda_val)
    discount = np.exp(-r * deltaT)

    # Set up S values (at maturity)
    Svals = np.zeros(2*N+1)
    Svals[0] = S0 * np.exp(N * deltaX)
    exp_dX = np.exp(-deltaX)

    for j in range(1, 2*N+1):
        Svals[j] = exp_dX * Svals[j-1]

    # Set up lattice and terminal values
    Pvals = np.zeros((2*N+1, N+1))

    for m in range(0, 2*N+1):
        Pvals[m, N] = max(K - Svals[m], 0)

    for j in range(N-1, -1, -1):
        for i in range(N-j, N+j+1):
            hold = discount * (p_u * Pvals[i-1, j+1] + p_m * Pvals[i, j+1] + p_d * Pvals[i+1, j+1])
            Pvals[i, j] = max(K - Svals[i], hold)

    price = Pvals[N, 0]
    computation_time = time.time() - start_time

    return price, computation_time

def TrinomialAmCall(S0, K, r, sigma, T, N, lambda_val=np.sqrt(3)):
    start_time = time.time()

    deltaT = T / N
    p_u, p_m, p_d, deltaX, lambda_used = TrinomialProbs(r, sigma, deltaT, lambda_val)
    discount = np.exp(-r * deltaT)

    # Set up S values (at maturity)
    Svals = np.zeros(2 * N + 1)
    Svals[0] = S0 * np.exp(N * deltaX)
    exp_dX = np.exp(-deltaX)

    for j in range(1, 2 * N + 1):
        Svals[j] = exp_dX * Svals[j - 1]

```

```

# Set up lattice and terminal values
Cvals = np.zeros((2 * N + 1, N + 1))

# Terminal payoff for a call
for k in range(0, 2 * N + 1):
    Cvals[k, N] = max(Svals[k] - K, 0)

# Backward induction
for j in range(N - 1, -1, -1):
    for i in range(N - j, N + j + 1):
        hold = discount * (p_u * Cvals[i - 1, j + 1] +
                             p_m * Cvals[i, j + 1] +
                             p_d * Cvals[i + 1, j + 1])
        # Early exercise condition for American call
        Cvals[i, j] = max(Svals[i] - K, hold)

price = Cvals[N, 0]
computation_time = time.time() - start_time

return price, computation_time

def RichardsonExtrapolation(P_N, P_2N, order=2):
    """
    ---Richardson-extrapolation-for-error-estimation
    ---P_N:-price-with-N-steps
    ---P_2N:-price-with-2N-steps
    ---order:-convergence-order-(typically-2-for-trinomial-trees)
    ---"""
    return (2**order * P_2N - P_N) / (2**order - 1)

def calculate_errors(prices, benchmark_price):
    """Calculate-absolute-error,-percentage-error,-and-standard-error"""
    abs_errors = [abs(price - benchmark_price) for price in prices]
    pct_errors = [abs_error / benchmark_price * 100 for abs_error in abs_errors]

    # Standard error of the mean (assuming prices are samples)
    if len(prices) > 1:
        std_error = np.std(prices, ddof=1) / np.sqrt(len(prices))
    else:
        std_error = 0

    return abs_errors, pct_errors, std_error

# Test parameters
S0 = 60
K = 65
sigma = 0.3
r = 0.03
T = 1.0

# Test different sample sizes
N_values = [2**6, 2**7, 2**8, 2**9, 2**10, 2**11, 2**12, 2**13]

print("====Convergence-Analysis-for-European-Call====")
print(f"{'N':>8}-{'Price':>10}-{'Abs-Error':>12}-{'Pct-Error(%)':>12}-{'Time(s)':>8}")
print("-" * 65)

eu_call_prices = []
eu_call_times = []
for N in N_values:
    price, comp_time = TrinomialEuCall(S0, K, r, sigma, T, N)
    eu_call_prices.append(price)
    eu_call_times.append(comp_time)

# Use Richardson extrapolation as benchmark for European call
richardson_benchmark_eu_call = RichardsonExtrapolation(eu_call_prices[-2], eu_call_prices[-1])

abs_errors, pct_errors, std_error = calculate_errors(eu_call_prices, richardson_benchmark_eu_call)

for i, N in enumerate(N_values):
    print(f"{'N':>8}-{'eu_call_prices[i]':>10.6f}-{'abs_errors[i]':>12.6f}-{'pct_errors[i]':>12.4f}-{'eu_call_times[i]':>8.6f}")

print(f"\nRichardson-Benchmark-European-Call:-{richardson_benchmark_eu_call:.6f}")
print(f"Standard-Error:-{std_error:.6f}")

#####
print("====Convergence-Analysis-for-European-Put====")
print(f"{'N':>8}-{'Price':>10}-{'Abs-Error':>12}-{'Pct-Error(%)':>12}-{'Time(s)':>8}")
print("-" * 65)

eu_put_prices = []
eu_put_times = []
for N in N_values:
    price, comp_time = TrinomialEuPut(S0, K, r, sigma, T, N)
    eu_put_prices.append(price)
    eu_put_times.append(comp_time)

```

```

# Use Richardson extrapolation as benchmark for European put
richardson_benchmark_eu_put = RichardsonExtrapolation(eu_put_prices[-2], eu_put_prices[-1])

abs_errors, pct_errors, std_error = calculate_errors(eu_put_prices, richardson_benchmark_eu_put)

for i, N in enumerate(N_values):
    print(f"{N}>8}-{eu_put_prices[i]:>10.6f}-{abs_errors[i]:>12.6f}-{pct_errors[i]:>12.4f}-{eu_put_times[i]:>8}")

print(f"\nRichardson-Benchmark-(European-Put):-{richardson_benchmark_eu_put:.6f}")
print(f"Standard-Error:-{std_error:.6f}")

#####
print("\n====-Convergence-Analysis-for-American-Call-====")
print(f"{N':>8}-{Price':>10}-{Abs-Error':>12}-{Pct-Error(%)':>12}-{Time(s)':>8}")
print("-" * 65)

am_call_prices = []
am_call_times = []
for N in N_values:
    price, comp_time = TrinomialAmCall(S0, K, r, sigma, T, N)
    am_call_prices.append(price)
    am_call_times.append(comp_time)

# Use Richardson extrapolation as benchmark for American call
richardson_benchmark_am_call = RichardsonExtrapolation(am_call_prices[-2], am_call_prices[-1])

abs_errors, pct_errors, std_error = calculate_errors(am_call_prices, richardson_benchmark_am_call)

for i, N in enumerate(N_values):
    print(f"{N}>8}-{am_call_prices[i]:>10.6f}-{abs_errors[i]:>12.6f}-{pct_errors[i]:>12.4f}-{am_call_times[i]:>8}")

print(f"\nRichardson-Benchmark-(American-Call):-{richardson_benchmark_am_call:.6f}")
print(f"Standard-Error:-{std_error:.6f}")

#####
print("\n====-Convergence-Analysis-for-American-Put-====")
print(f"{N':>8}-{Price':>10}-{Abs-Error':>12}-{Pct-Error(%)':>12}-{Time(s)':>8}")
print("-" * 65)

am_put_prices = []
am_put_times = []
for N in N_values:
    price, comp_time = TrinomialAmPut(S0, K, r, sigma, T, N)
    am_put_prices.append(price)
    am_put_times.append(comp_time)

# Use Richardson extrapolation as benchmark for American put
richardson_benchmark_am_put = RichardsonExtrapolation(am_put_prices[-2], am_put_prices[-1])

abs_errors, pct_errors, std_error = calculate_errors(am_put_prices, richardson_benchmark_am_put)

for i, N in enumerate(N_values):
    print(f"{N}>8}-{am_put_prices[i]:>10.6f}-{abs_errors[i]:>12.6f}-{pct_errors[i]:>12.4f}-{am_put_times[i]:>8}")

print(f"\nRichardson-Benchmark-(American-Put):-{richardson_benchmark_am_put:.6f}")
print(f"Standard-Error:-{std_error:.6f}")

def calculate_errors(prices, benchmark_price):
    """Calculate absolute error, percentage error, and standard error"""
    abs_errors = [abs(price - benchmark_price) for price in prices]
    pct_errors = [abs_error / benchmark_price * 100 for abs_error in abs_errors]

    if len(prices) > 1:
        std_error = np.std(prices, ddof=1) / np.sqrt(len(prices))
    else:
        std_error = 0

    return abs_errors, pct_errors, std_error

# Test parameters
S0 = 60
K = 65
sigma = 0.3
r = 0.03
T = 1.0

N_values = [2**6, 2**7, 2**8, 2**9, 2**10, 2**11, 2**12, 2**13]

#####
print("\n====-Convergence-Analysis-for-European-Call-====")
print(f"{N':>8}-{Price':>10}-{Abs-Error':>12}-{Pct-Error(%)':>12}-{Time(s)':>8}")
print("-" * 65)

eu_call_prices = []
eu_call_times = []
for N in N_values:

```

```

price, comp_time = TrinomialEuCall(S0, K, r, sigma, T, N)
eu_call_prices.append(price)
eu_call_times.append(comp_time)

if len(eu_call_prices) >= 3:
    P_N = eu_call_prices[-3]
    P_2N = eu_call_prices[-2]
    P_4N = eu_call_prices[-1]

    R1_call = RichardsonExtrapolation(P_N, P_2N)
    R2_call = RichardsonExtrapolation(P_2N, P_4N)
    higher_order_estimate_eu_call = (4 * R2_call - R1_call) / 3

    print("\nEuropean Call Higher-Order Richardson:")
    print(f"Second-order Richardson (2^11, 2^12): {R1_call:.6f}")
    print(f"Second-order Richardson (2^12, 2^13): {R2_call:.6f}")
    print(f"Third-order Richardson estimate: {higher_order_estimate_eu_call:.6f}")

    abs_errors, pct_errors, std_error = calculate_errors(eu_call_prices, higher_order_estimate_eu_call)

    for i, N in enumerate(N_values):
        print(f"{N:>8} {eu_call_prices[i]:>10.6f} {abs_errors[i]:>12.6f} {pct_errors[i]:>12.4f} {eu_call_times[i]:>12.4f}")

    print(f"\nRichardson Benchmark (European Call): {higher_order_estimate_eu_call:.6f}")
    print(f"Standard Error: {std_error:.6f}")

#####
print("==== Convergence Analysis for European Put ===")
print(f"{'N':>8} {'Price':>10} {'Abs-Error':>12} {'Pct-Error(%)':>12} {'Time(s)':>8}")
print("_" * 65)

eu_put_prices = []
eu_put_times = []
for N in N_values:
    price, comp_time = TrinomialEuPut(S0, K, r, sigma, T, N)
    eu_put_prices.append(price)
    eu_put_times.append(comp_time)

if len(eu_put_prices) >= 3:
    P_N = eu_put_prices[-3]
    P_2N = eu_put_prices[-2]
    P_4N = eu_put_prices[-1]

    R1_put = RichardsonExtrapolation(P_N, P_2N)
    R2_put = RichardsonExtrapolation(P_2N, P_4N)
    higher_order_estimate_eu_put = (4 * R2_put - R1_put) / 3

    print("\nEuropean Put Higher-Order Richardson:")
    print(f"Second-order Richardson (2^11, 2^12): {R1_put:.6f}")
    print(f"Second-order Richardson (2^12, 2^13): {R2_put:.6f}")
    print(f"Third-order Richardson estimate: {higher_order_estimate_eu_put:.6f}")

    abs_errors, pct_errors, std_error = calculate_errors(eu_put_prices, higher_order_estimate_eu_put)

    for i, N in enumerate(N_values):
        print(f"{N:>8} {eu_put_prices[i]:>10.6f} {abs_errors[i]:>12.6f} {pct_errors[i]:>12.4f} {eu_put_times[i]:>12.4f}")

    print(f"\nRichardson Benchmark (European Put): {higher_order_estimate_eu_put:.6f}")
    print(f"Standard Error: {std_error:.6f}")

#####
print("==== Convergence Analysis for American Call ===")
print(f"{'N':>8} {'Price':>10} {'Abs-Error':>12} {'Pct-Error(%)':>12} {'Time(s)':>8}")
print("_" * 65)

am_call_prices = []
am_call_times = []
for N in N_values:
    price, comp_time = TrinomialAmCall(S0, K, r, sigma, T, N)
    am_call_prices.append(price)
    am_call_times.append(comp_time)

if len(am_call_prices) >= 3:
    P_N = am_call_prices[-3]
    P_2N = am_call_prices[-2]
    P_4N = am_call_prices[-1]

    R1_am_call = RichardsonExtrapolation(P_N, P_2N)
    R2_am_call = RichardsonExtrapolation(P_2N, P_4N)
    higher_order_estimate_am_call = (4 * R2_am_call - R1_am_call) / 3

    print("\nAmerican Call Higher-Order Richardson:")
    print(f"Second-order Richardson (2^11, 2^12): {R1_am_call:.6f}")
    print(f"Second-order Richardson (2^12, 2^13): {R2_am_call:.6f}")
    print(f"Third-order Richardson estimate: {higher_order_estimate_am_call:.6f}")

    abs_errors, pct_errors, std_error = calculate_errors(am_call_prices, higher_order_estimate_am_call)

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for i, N in enumerate(N_values):
    print(f"{N:>8}- {am_call_prices[i]:>10.6f}- {abs_errors[i]:>12.6f}- {pct_errors[i]:>12.4f}- {am_call_times[i]:>10.6f}")

print(f"\nRichardson-Benchmark-(American-Call):- {higher_order_estimate_am_call:.6f}")
print(f"Standard-Error:- {std_error:.6f}")

#####
print("\n==== Convergence-Analysis-for-American-Put====")
print(f"{N':>8}- {'Price':>10}- {'Abs-Error':>12}- {'Pct-Error(%)':>12}- {'Time(s)':>8}")
print("_" * 65)

am_put_prices = []
am_put_times = []
for N in N_values:
    price, comp_time = TrinomialAmPut(S0, K, r, sigma, T, N)
    am_put_prices.append(price)
    am_put_times.append(comp_time)

if len(am_put_prices) >= 3:
    P_N = am_put_prices[-3]
    P_2N = am_put_prices[-2]
    P_4N = am_put_prices[-1]

    R1_am_put = RichardsonExtrapolation(P_N, P_2N)
    R2_am_put = RichardsonExtrapolation(P_2N, P_4N)
    higher_order_estimate_am_put = (4 * R2_am_put - R1_am_put) / 3

    print("\nAmerican-Put-Higher-Order-Richardson:")
    print(f"Second-order-Richardson-(2^11,-2^12):- {R1_am_put:.6f}")
    print(f"Second-order-Richardson-(2^12,-2^13):- {R2_am_put:.6f}")
    print(f"Third-order-Richardson-estimate:- {higher_order_estimate_am_put:.6f}")

abs_errors, pct_errors, std_error = calculate_errors(am_put_prices, higher_order_estimate_am_put)

for i, N in enumerate(N_values):
    print(f"{N:>8}- {am_put_prices[i]:>10.6f}- {abs_errors[i]:>12.6f}- {pct_errors[i]:>12.4f}- {am_put_times[i]:>10.6f}")

print(f"\nRichardson-Benchmark-(American-Put):- {higher_order_estimate_am_put:.6f}")
print(f"Standard-Error:- {std_error:.6f}")

```